1. Introduction

Women's safety in public spaces remains a pressing concern, particularly in locations where incidents of harassment recur at specific places or during particular time periods. Existing safety applications largely focus on emergency assistance and incident reporting. On their own, however, incident records do not equip authorities to understand recurring patterns, identify areas needing attention, or make informed patrolling decisions.

This project proposes an integrated web-based and mobile-based safety system that converts women-safety incident data into actionable information for both authorities and users. Incidents are analysed by location, time, frequency and recurrence to identify areas where they are concentrated, and these areas are represented as geographical hotspots through interactive maps and visualizations.

A central component of the system is an interactive dashboard for police authorities. An officer such as a DSP can filter incident information by geographical region — for instance, selecting South Bangalore automatically updates the charts, statistics, maps and hotspot information for that region allowing the safety situation of a specific area to be examined without manually sifting through large volumes of incident data.

The system also supports patrolling decisions by highlighting locations with a higher concentration or recurrence of incidents, so that patrolling teams can direct their attention to areas that genuinely require intervention rather than deploying resources without adequate supporting information.

An automated daily alert mechanism forms another key component. By analysing the previous day's incident history, the system generates alerts that draw attention to areas with new incidents or recurring patterns, helping authorities identify locations that call for increased vigilance and preventive patrolling.

Alongside the authority dashboard, a mobile application delivers safety-oriented functionality to end users. Phase-I focuses on building this core application and its foundational features, with the architecture kept flexible for future enhancement. Once the initial version is complete, it can be demonstrated to relevant stakeholders so that further requirements can be incorporated in subsequent phases.

The system will initially be built using available regional incident data, with an architecture designed to extend to other cities and regions. Taken together, the intended flow of the system is:

Incident Data → Analysis → Hotspot Identification → Interactive Dashboard → Daily Alerts → Patrolling Support → Preventive Intervention

Fig. No : 1.1 Flow of the system

Within this flow, the mobile application is the user-facing component, while the dashboard and alert system together form the authority-facing component. The overall objective is to shift public safety from a purely reactive posture to a data-driven, preventive approach in which incident information supports timely intervention and the effective use of patrolling resources.

2. Literature Survey

2.1 Existing Women Safety Applications

Existing women-safety applications offer features such as SOS alerts, emergency contact notification, location sharing and emergency assistance. These are designed primarily to support an individual during an emergency, and generally provide limited analytical support to authorities for understanding incident patterns across multiple locations.

2.2 Crime Mapping and Geospatial Analysis

Geospatial analysis is widely used to represent incidents according to their geographical location. Mapping incidents lets authorities identify areas of concentration and provides a visual sense of spatial patterns, though static maps often lack the interaction, filtering and real-time decision support that operational use requires.

2.3 Hotspot Identification

Hotspot identification techniques determine locations where incidents cluster. Data analysis and clustering approaches group incidents by geographical distribution, helping authorities identify areas that need greater attention and enabling more focused resource allocation.

2.4 Public-Safety Dashboards

Interactive dashboards present large volumes of information in an accessible form. Filtering lets decision-makers examine specific locations, time periods and incident categories, and dynamic dashboards that update charts and maps based on selected parameters are particularly useful for authority-level monitoring.

2.5 Automated Alerts and Preventive Intervention

Automated notification systems give authorities timely information about newly reported incidents or areas needing attention. Combined with historical analysis, daily alerts can highlight recurring or concentrated incident locations and support preventive decision-making.

2.6 Research Gap

Existing solutions tend to focus on either individual emergency assistance or incident reporting and visualization in isolation. There remains a need for an integrated platform that combines incident analysis, geographical hotspot identification, interactive filtering, automated alerts, patrolling support and user-oriented safety functionality within a single system.

This project addresses that gap by bringing these components together, converting incident data into actionable information that helps authorities identify areas requiring attention while also providing safety-oriented functionality directly to users.

3. Problem Statement, Aim and Objectives

3.1 Problem Statement

Existing women-safety systems mainly provide emergency assistance and incident-reporting facilities. Authorities, however, need stronger analytical support to understand where incidents are repeatedly occurring, identify high-risk locations, monitor specific geographical regions, and determine where patrolling resources should be focused.

There is therefore a need for an integrated system that analyses incident information, identifies geographical hotspots, provides interactive dashboards and automated alerts, and supports data-driven, preventive patrolling while also offering basic real-time safety functionality to users.

3.2 Aim

To develop an integrated incident analysis and women-safety system that identifies high-risk locations, provides interactive dashboards, generates automated alerts, and supports data-driven patrolling and preventive intervention.

3.3 Objectives

1. To collect and organize women-safety incident data.
2. To analyse incidents based on location, time, frequency and recurrence.
3. To identify geographical areas where incidents are concentrated.
4. To visualize high-risk areas using interactive maps.
5. To develop an interactive dashboard for police authorities.
6. To provide geographical filtering for analysing specific regions.
7. To dynamically update dashboard charts, statistics, maps and hotspot information according to selected filters.
8. To generate automated daily alerts based on previous-day incident information.
9. To provide information that supports patrolling teams in identifying areas requiring greater attention.
10. To provide incident-based justification for identified high-risk areas.
11. To develop a basic mobile application with safety-oriented functionality.
12. To design the system so that it can be extended to different cities and regions.

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4. SDG & ESG Compliance

SDG vs ESG — what's the difference?

SDG (Sustainable Development Goals): 17 global goals set by the UN (e.g., Clean Water, Good Health, Climate Action). You pick ONE goal your project supports, along with a specific target/indicator under it.

ESG (Environmental, Social, Governance): A broader framework for how responsibly a project is built and run. You describe your project's contribution under all THREE categories — Environmental, Social, and Governance.

In short: SDG = "which global goal does my project support" and ESG = "how sustainably/responsibly is my project designed, built, and governed." A project usually maps to ONE SDG, but should address ALL THREE ESG categories.

A. Section A — SDG Compliance

i. Project title: Incident Analysis and Hotspot Identification for Security Provision Against Women Harassment in Public Spaces

ii. Identified SDG (Goal No. & Name):

Sustainable Development Goal (SDG): SDG 5 – Gender Equality

iii. Target & Indicator under this SDG:

Target 5.2 – Eliminate all forms of violence against all women and girls in public and private spheres, including trafficking, sexual exploitation, and other forms of violence.

Indicator 5.2.2 – Proportion of women and girls subjected to sexual violence by persons other than an intimate partner, by age and place of occurrence.

iv. Contribution of the Project towards the identified SDG:

This project aligns with SDG 5 by addressing women's safety in public spaces and supporting preventive measures against harassment. Its contribution can be summarised as follows:

- The system analyses women-safety incidents and identifies locations where incidents are concentrated or repeatedly occurring.
- Hotspot visualization and automated alerts give authorities information that can support preventive patrolling and timely intervention.
- The mobile application delivers safety-oriented functionality to users, contributing to safer participation in public spaces.

v. Supporting Documents / Evidence:

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B. Section B — ESG Compliance

ESG Category	Project Contribution	Supporting Documents
Environmental (E)	The system uses digital reporting and storage instead of depending on paper records. This helps reduce paper usage and makes incident records easier to maintain digitally.	System design, database design, web/mobile application screenshots
Social (S)	The system helps authorities and users understand where and when women's safety incidents occur more frequently. By identifying recurring incidents and high-risk areas, it supports better patrolling and safety planning. Interactive maps and dashboards make the information easier to understand and help authorities make more informed decisions based on actual incident data.	Project synopsis, hotspot maps, authority dashboard, mobile application screenshots, SDG alignment
Governance (G)	The system protects incident information by using authentication and role-based access. This ensures that only authorized users can access the data and helps maintain the privacy and security of sensitive information.	System architecture, authentication module, access-control design, security documentation

Signature of Student **Signature of Project Guide**

5. Methodology

The proposed system will be developed through the following stages.

5.1 Data Collection

Incident data will be collected from available datasets and, where applicable, authorized sources. Relevant attributes such as incident location, date, time, category and frequency will be maintained.

5.2 Data Pre-processing

The collected data will be cleaned and prepared for analysis. Missing, duplicate and inconsistent records will be handled to improve dataset quality.

5.3 Incident Analysis

The processed data will be analysed to identify:

- Frequently occurring locations
- Time-wise incident patterns
- Area-wise incident frequency
- Repeated incidents
- Geographical concentration of incidents

5.4 Geospatial Analysis

Incident locations will be represented geographically using interactive maps, allowing authorities to visually understand the distribution of incidents across different regions.

5.5 Hotspot Identification

Suitable data-analysis and clustering techniques will be used to identify areas where incidents are concentrated. Identified locations will be represented using graded levels of concern:

Low Risk → Moderate Risk → High Risk

Fig. No: 5.5.1 Grade levels

High-risk areas will be highlighted as hotspots.

5.6 Interactive Dashboard Development

An interactive dashboard will be developed for police authorities, containing:

- Incident statistics
- Charts
- Interactive maps
- Hotspots
- Time-based analysis
- Region-based filtering
- Area-wise information

When a particular region is selected, all relevant dashboard components will update automatically to reflect that selection.

5.7 Automated Daily Alert Generation

The system will analyse the previous day's incident information and generate a daily summary that can highlight:

- Areas where incidents were reported
- Areas with repeated incidents
- High-risk locations
- Locations requiring increased attention

5.8 Patrolling Support

The system will provide information about high-risk locations to support patrolling decisions, backed by incident history and hotspot information, so that authorities have a data-based justification for focusing resources in a particular area.

5.9 Mobile Application

A mobile application will provide the user-facing safety component. Phase-I focuses on the basic application and core safety functionality, with additional features to be incorporated in future phases.

5.10 Testing and Validation

The developed modules will be tested to verify:

- Correct data processing
- Accurate dashboard filtering
- Dynamic visualization
- Hotspot display
- Alert generation
- Mobile application functionality

6. Modules Identified

Module 1: Incident Data Management

This module encompasses all the processes involved with gathering and managing the information regarding the women-safety incidents. The data reports of the incidents will be stored in a structured manner including the location, date, time, nature, and any other pertinent information. The information will be pre-processed and prepared for analysis by the system while leaving the option for adding new data or updating the existing information.

Module 2: Incident Analysis

This module will involve the analysis of the data gathered pertaining to incidents and their location, time, and date of occurrence. The incidents will be studied on the basis of the location, time, and frequency, and areas will be identified where the incidents are repetitive or frequent.

Module 3: Hotspot Identification

This module is for identifying hotspots on the basis of the information analyzed and processed above. The gathered data will be used to find out probabilities of safety issues and will be categorized on the basis of risk levels. This information will then be presented in the form of maps showing hotspots with varying risk levels.

Module 4: Authority Dashboard

For the authorities, this system will have a dashboard that will serve as a repository for all the information and will provide summary statistics that may be filtered on the basis of regions or any other factors. The dashboard will include visualizations such as charts, graphs, maps, hotspots, and time-series analysis with options to sort and view the required information.

Module 5: Automated Alert System

Authorities will be provided with automated alerts and summaries of incidents that occur on a daily basis. The data can be analyzed to create alerts and daily summaries of the most dangerous hotspots after assessing the previous day's data without the need to go through all the data.

Module 6 : Patrolling Support

The patrolling support will be the utilization of the data processed and gathered to assess the patrolling needs and requirements. In this way, the authorities will be able to make informed decisions regarding patrolling based on the data of frequent and dangerous hotspots instead of making decisions based on assumptions.

Module 7 : Mobile Safety Application

The general public can be given a mobile application with safety features as the front-end application. This application will preferably be intuitive and have basic location-based services, real-time safety support, and emergency features and services. These features can be adjusted and finalized in regards to the applicable features and the feasibility of the application.

Module 8 : Administration and Security

The administration and security entails the aspect of the system administration and security protocols that will be utilized to store the data safely and securely while also restricting the data from unauthorized access. The data can be protected, and access can be given to the relevant authorities only after authentication. The administration will take care of the ethical aspects of the data and will ensure that sensitive data is handled responsibly. The authorities can be provided with different levels of access according to their roles and responsibilities in the system.

7. Tools and Technologies Identified

Table No: 7.1 Tools and technologies

Component	**Technology**
Frontend	React.js, HTML, CSS, JavaScript
Backend	Python, FastAPI
Database	MySQL / MongoDB
Data Analysis	Python, Pandas, NumPy
Machine Learning	Scikit-learn
Data Visualization	Plotly / suitable visualization libraries
Geospatial Visualization	Interactive map-based libraries
Mobile Application	Android / suitable cross-platform framework
API Development	REST API
Version Control	Git and GitHub
Development Environment	Visual Studio Code / Android Studio
Deployment	Cloud platform as required

8. System Model — Process Flow

The diagrams below present a simple model of how the system operates: the first shows the overall process flow from incident data to preventive intervention, and the second shows how the major components fit together at an architectural level.

8.1 Process Flow Model

Incident data moves through a shared analysis pipeline that produces the interactive dashboard, daily alerts and patrolling support for authorities, while the mobile application delivers real-time safety functionality directly to users.

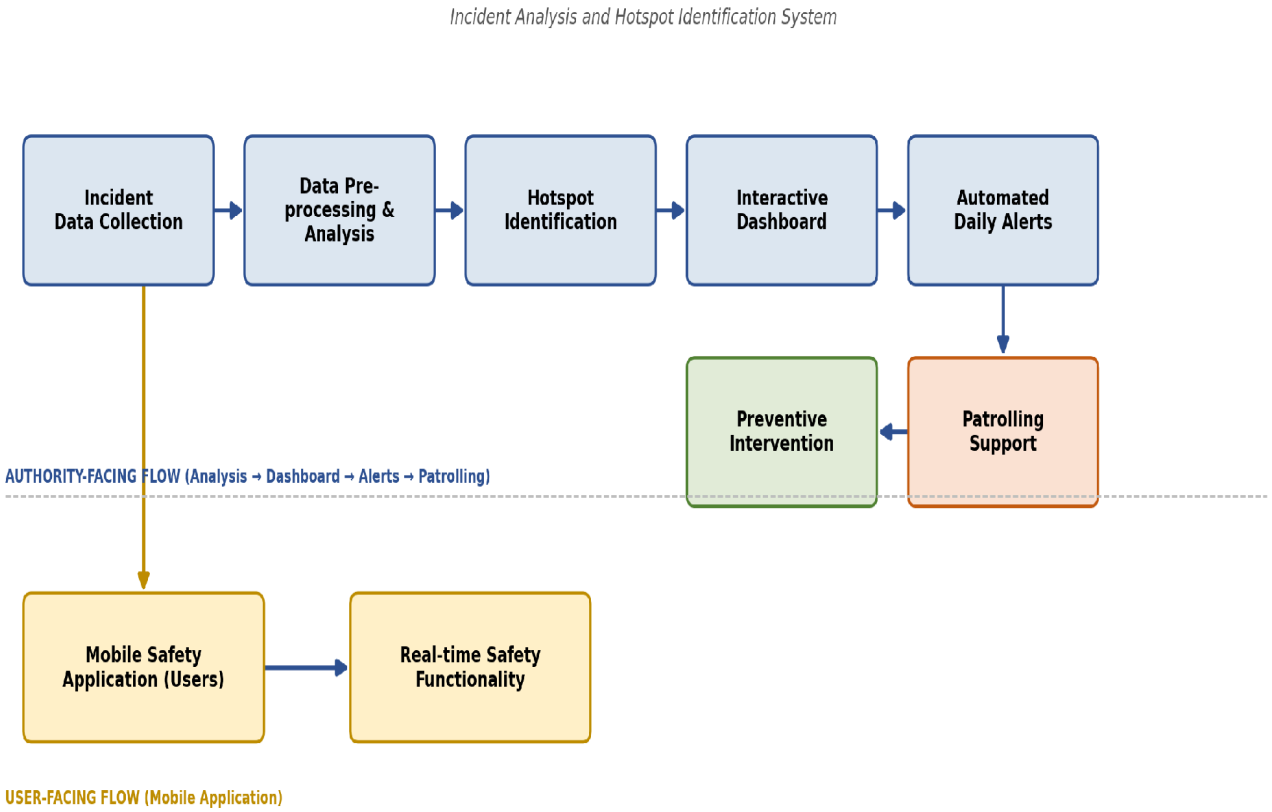


Fig. 8.1 — Process flow from incident data to preventive intervention.

8.2 System Architecture Model

Data collected from incident sources is cleaned and stored, then passed through the analysis engine for hotspot and risk classification. Results are exposed through a REST API layer that feeds both the authority dashboard and the mobile application, with an administration and security layer governing access throughout.

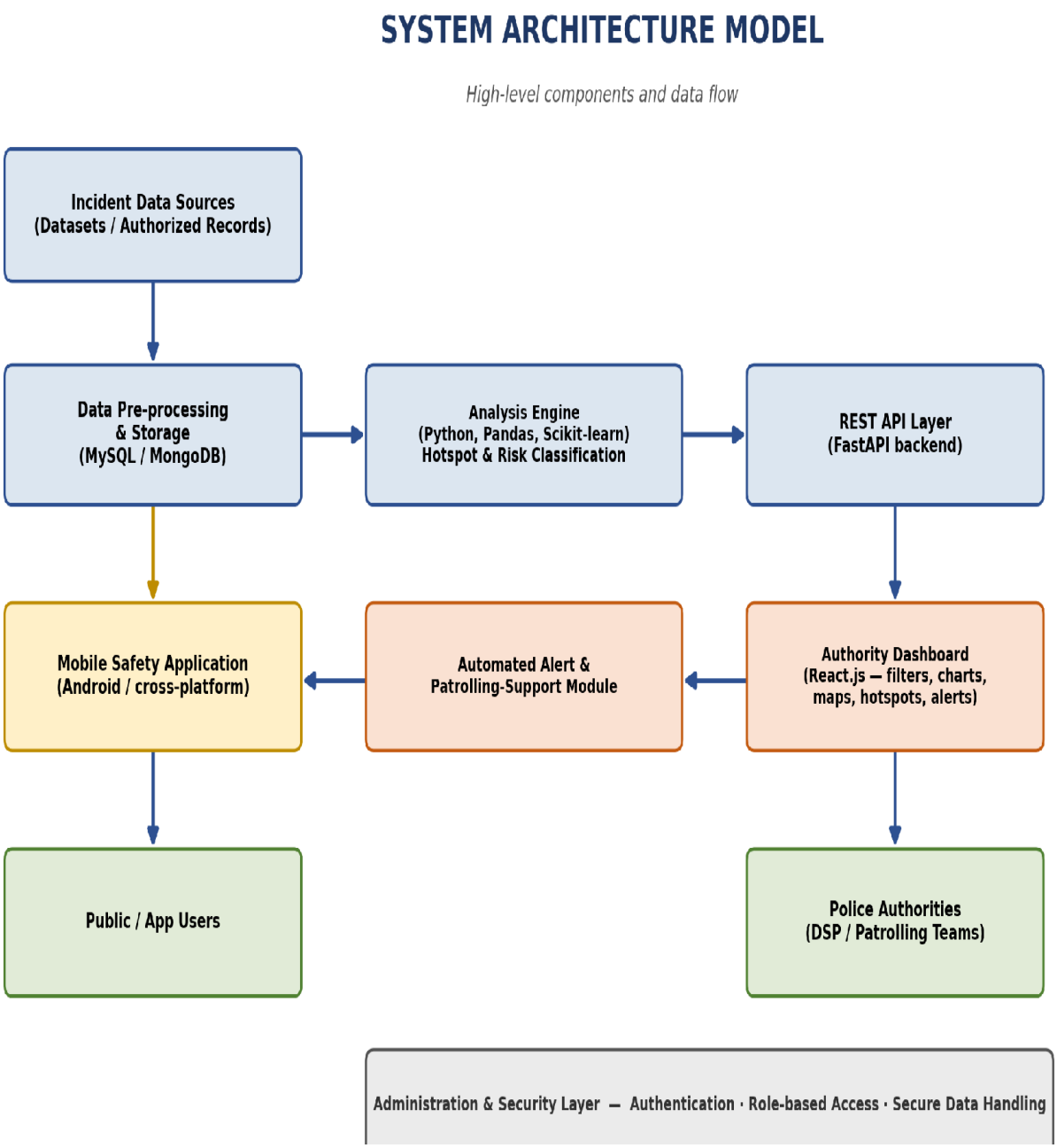


Fig. 8.2 — High-level system architecture and data flow.

9. Timeline using Gantt Chart

Table No: 9.1 Gantt Chart

Activities	**Week 1**	**Week 2**	**Week 3**	**Week 4**
Problem Identification & Requirement Analysis	✓			
Literature Survey	✓	✓		
Dataset Collection & Preparation		✓		
Data Pre-processing & Analysis		✓	✓	
Hotspot Identification			✓	
Authority Dashboard Development			✓	✓
Mobile Application — Basic Version			✓	✓
Automated Alerts & Patrolling Support				✓
Testing & Validation				✓
Documentation & Demonstration				✓

Phase-I Expected Deliverable

At the end of Phase-I, the project is expected to deliver a basic functional prototype comprising an interactive authority dashboard, incident analysis, geographical hotspot visualization, region-based filtering, automated daily alerts, initial patrolling-support functionality, and a basic mobile safety application.

The application structure will also accommodate future enhancements. Once the basic version is developed, it can be demonstrated to relevant stakeholders to understand their practical requirements and incorporate appropriate improvements in subsequent phases.